



EB-GS2962

Design Guide

Version	ECR	Date	Changes and / or Modifications
0	151474	February 2009	New document.

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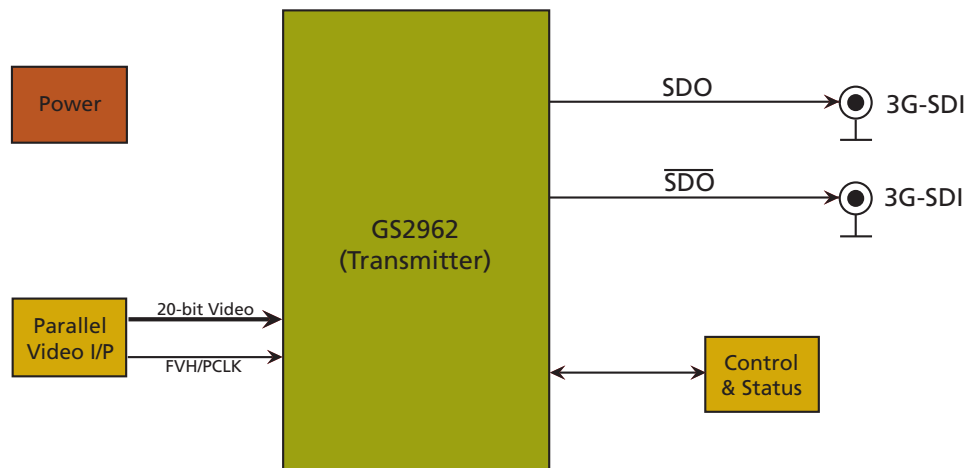
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Overview

Together with the EB-GS2962 Evaluation Board, this document serves as a reference for designing with the GS2962, a Gennum 3Gb/s SDI Transmitter. This document contains two main areas of focus:

1. Considerations in the schematic design.
2. Recommended PCB layout practices when designing with the GS2962.

The figure below shows a block diagram overview of the features and the functions of EB-GS2962.



Block Diagram of the EB-GS2962

1. Power

1.1 Power Supply

The GS2962 requires a stable ($\pm 5\%$), low noise power supply. Ideally, these voltages are provided by linear voltage regulators.

The GS2962 requires two voltages; +3.3V and +1.2V, with an optional +1.8V for the digital I/Os.

The +1.2V powers the digital core as well as the analog portions of the chip, such as the PLL components. The +3.3V powers the on-chip cable driver, SDI input buffer, other sensitive analog circuitry and digital I/O (the digital I/Os can be powered either by +1.8V or +3.3V). For optimal performance, it is essential that these supplies are isolated to avoid external noise coupling. It is recommended to use separate power planes for those supplies, as well as two separate ground planes, analog and digital.

These supplies can be broken down into five groups; +1.2VA/A_GND, +3.3VA/A_GND, CD_VDD/A_GND, +1.2V/GND and IO_VDD/GND.

See [Figure 1-1: GS2962 power supply filtering and decoupling](#).

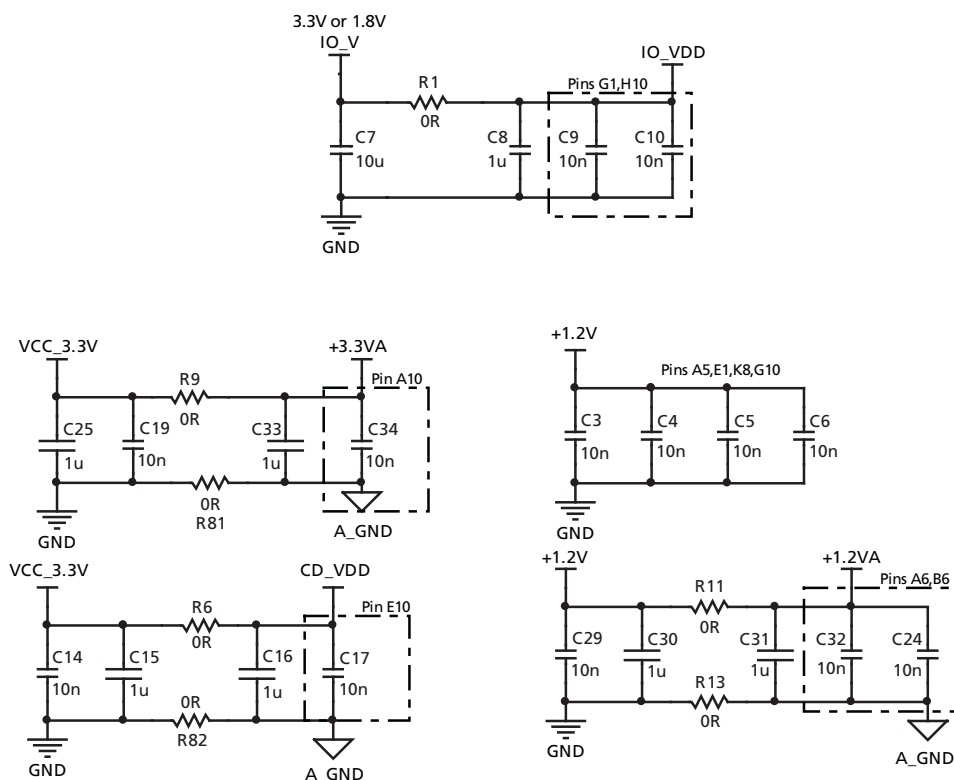


Figure 1-1: GS2962 power supply filtering and decoupling

The following guidelines are suggested when designing power for the GS2962:

- Use coupled power and ground planes i.e. use minimum spacing between power and ground planes. Power and ground planes form a natural capacitor, which will increase the capacitance
- Do not overlap power planes. If it is unavoidable, different power planes should be isolated from each other with the ground plane between them
- Power and ground planes should be placed near the component side, which will reduce inductance of vias
- If possible, use multiple vias to connect components to power supply.
- Use low ESL, low ESR capacitors

Isolate the power plane and the ground plane from the main plane in a 'moat'. Gennum recommends that power and ground connections to this 'island' made through 0Ω resistors, which are decoupled on both sides. In the case where better filtering is needed, 0Ω resistors can be replaced with an inductor or ferrite beads. Use multiple vias and copper pours to connect those components to the main plain and isolated planes. Place at least 1μF capacitor on the entrance of the power plane. If possible, running high-speed traces across a moat should be avoided. See [Figure 1-2](#).

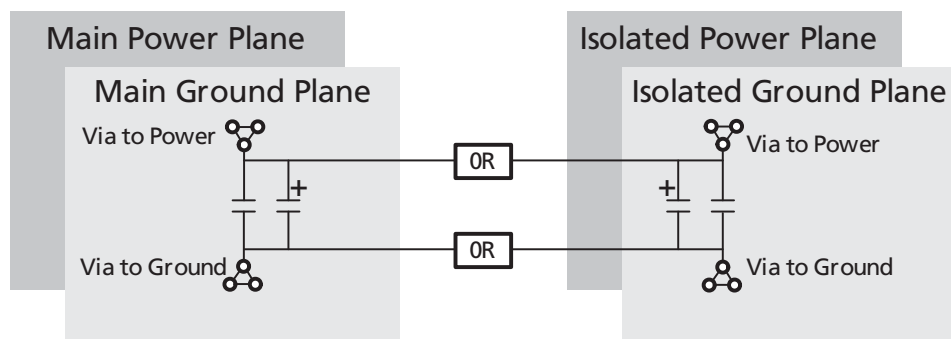


Figure 1-2: Isolated powers in a moat

1.2 Power Filtering and Decoupling

It is recommended to decouple each power supply pin with a 10nF capacitor. Decoupling capacitors should be connected between a power supply and the related ground, see [Figure 1-2](#).

All decoupling and filtering ceramic capacitors should be placed as close as possible to the related pins of the GS2962. Please see [Figure 1-1](#).

Ideally, the bypass capacitors should be physically located between the power plane and the power pin. Current should flow from the power plane to the capacitor to the power pin.

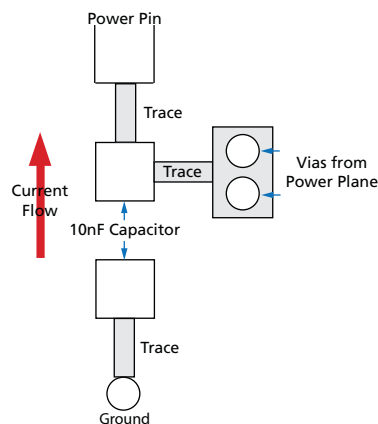


Figure 1-3: Power filtering and decoupling

For the power pins of the outside rows of a BGA, the decoupling capacitors should be placed on the top layer. For the power pins of the inner rows of a BGA, the decoupling capacitors can be placed on the bottom layer, close to the related power pin.

1.3 VCO Decoupling

To minimize jitter on SDI outputs, it is particularly important to maintain low noise on the VCO power supply. It is recommended to connect the VCO_VDD pin to the 1.2V

analog power plane through the RC filter shown in Figure 1-4. The RC components should be placed as close as possible to the GS2962, and away from noise sources.

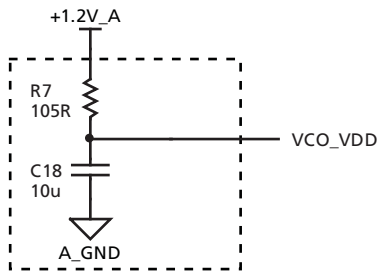


Figure 1-4: VCO Decoupling

2. SDI Input and Output

2.1 SDI Output

The GS2962 contains a SMPTE compliant cable driver, designed to drive one or two 75Ω coaxial cables. The SMPTE specifications require that SDI outputs meet an Output Return Loss (ORL) specification. Figure 2-1 shows the SDI output interface with ORL components. NOTE: The values for the ORL components are application and PCB-layout specific, but the recommendations below can be a good starting point for designing a SDI interface.

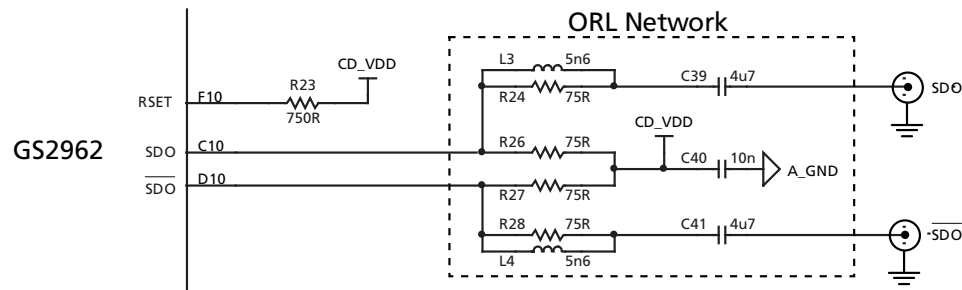


Figure 2-1: SDI interface

Special attention must be paid to component layout when designing 3Gb/s SDI outputs. An FR-4 dielectric can be used, however, the following layout techniques should be used to optimise performance:

- Use controlled-impedance transmission lines, referenced to the analog ground only
- The PCB trace should be closely matched to the SMT component width to minimize reflections due to changes in trace impedance
- Use anti-pads (cut outs) for output components to minimize parasitic capacitance. The anti-pad should extend out to the edge of the second pad of this passive component
- Use surface mounted in-line BNC connectors to eliminate a transmission line stub caused by a BNC mounting

- Curve high-speed traces to minimize impedance variations due to change of PCB trace
- Place the ORL circuitry and the BNCs close to the GS2962
- Ensure that the traces and the component placements are symmetrical
- Do not use silk-screen on these traces to ensure accurate impedance
- Use copper pour to isolate traces and components from surrounding noise sources
- The trace from the BNC to the capacitor should be of equal width to the pad of the capacitor

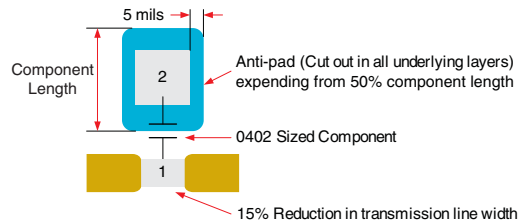


Figure 2-2: Anti-pad for Shunt Component on Transmission Lines

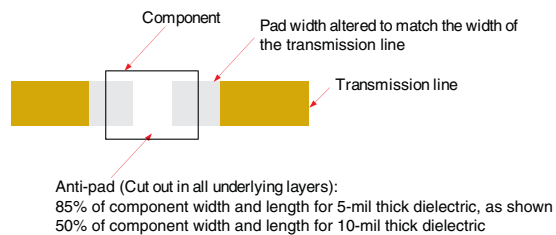


Figure 2-3: Anti-pad for Serial Components on Transmission Lines

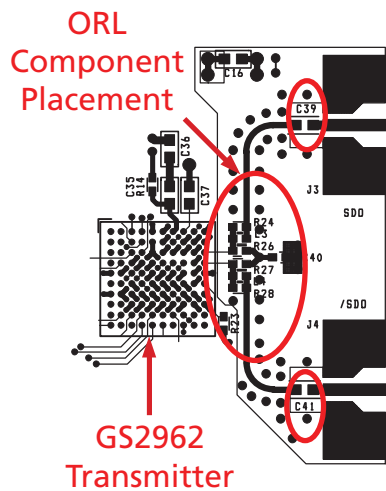


Figure 2-4: ORL Component Placement

2.2 Output Amplitude (RSET)

The output amplitude of the GS2962 can be adjusted by changing the value of the RSET resistor as shown in Table 2-1. For a 800mV_{p-p} output a value of 750Ω is required. A ±1% SMT resistor should be used.

The RSET resistor is part of the high speed output circuit of the GS2962. The resistor should be placed as close as possible to the RSET pin. In addition, an anti-pad should be used underneath the resistor.

Table 2-1: RSET Resistor Values vs. Output Swing

RSET Resistor Values (Ω)	Output Swing (mV _{p-p})
	608
824	734
750	800
680	884

3. Parallel Inputs (Clock and Data)

The GS2962 features a Dual Data Rate (DDR) input, in which case, the maximum data rate is 297Mb/s.

The primary consideration with clock and input data traces is to route them in a manner that reduces capacitive and inductive crosstalk, while maintaining equivalent length.

To minimize the crosstalk, the clock traces must be at least 3x the trace width away from adjacent signals, or the clock trace can be isolated by low impedance reference (power or ground). If possible try to avoid vias on the clock trace.

To get a reasonable timing margin, keep the routing lengths of the bus signals and synchronous signals to approximately the same length (<±0.5" tolerance) with the GS2962_PCLK in average length. If Dual Data Rate output is not required, or the length of the traces is not more than 2", trace length matching is not required.

NOTE: The GS2962 I/O cells are not 3.3V tolerant when the I/O supply voltage is set to 1.8V.

4. Loop filter and bandgap inputs

Figure 4-1 shows recommended loop filter and bandgap components for GS2962 PLL.

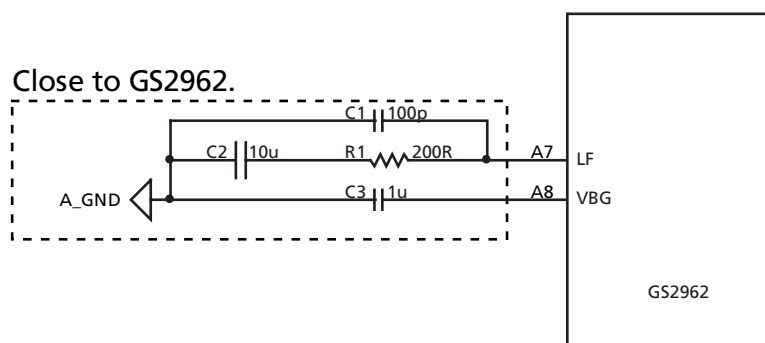


Figure 4-1: Recommended loop filter and bandgap components

Since these are noise sensitive inputs, place the components close to the GS2962 and avoid routing of the other digital signals under them or close to them.

DOCUMENT IDENTIFICATION DESIGN GUIDE

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